Macroeconomics: Problem Set 1

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Exercise 1

Data sources

All series were downloaded in `R` from the Federal Reserve Bank of St. Louis database (FRED) using `symbols <- c( "GDPC1", "PCECC96", "NAEXKP01DEQ661S", "NAEXKP02DEQ661S" ) getSymbols(symbols, src = "FRED")`.

For the United States, the series used are *Real Gross Domestic Product* (GDPC1)(U.S. Bureau of Economic Analysis, 2026a) and *Real Personal Consumption Expenditures* (PCECC96) (U.S. Bureau of Economic Analysis, 2026b).

For Germany, the series used are *National Accounts: GDP by Expenditure: Constant Prices: Gross Domestic Product: Total for Germany* (NAEXKP01DEQ661S) (Organisation for Economic Co-operation and Development, 2026a) and *National Accounts: GDP by Expenditure: Constant Prices: Private Final Consumption Expenditure for Germany* (NAEXKP02DEQ661S) (Organisation for Economic Co-operation and Development, 2026b).

Task 1

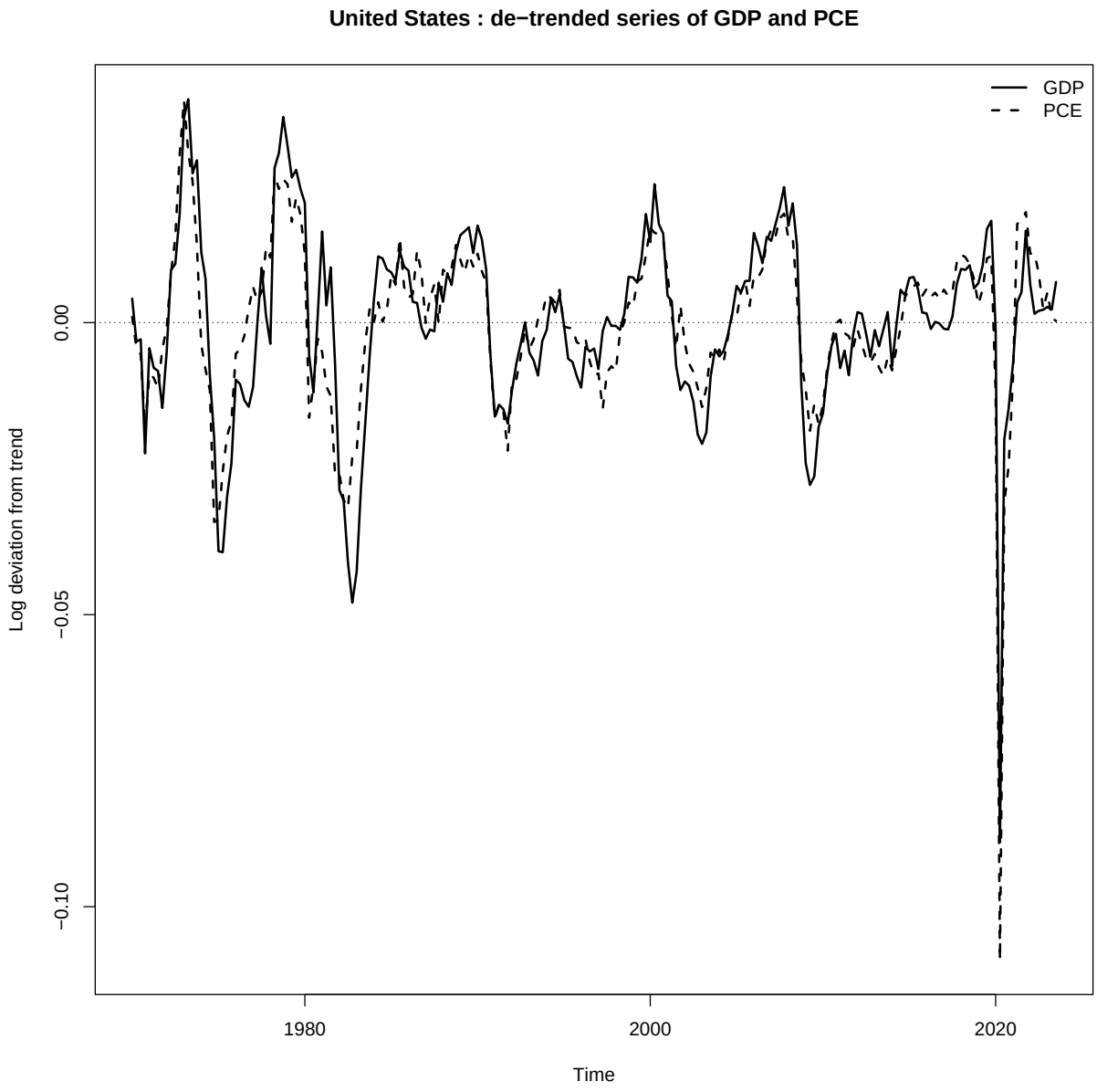

Figure 1: De-trended GDP and PCE in the United States using the HP filter

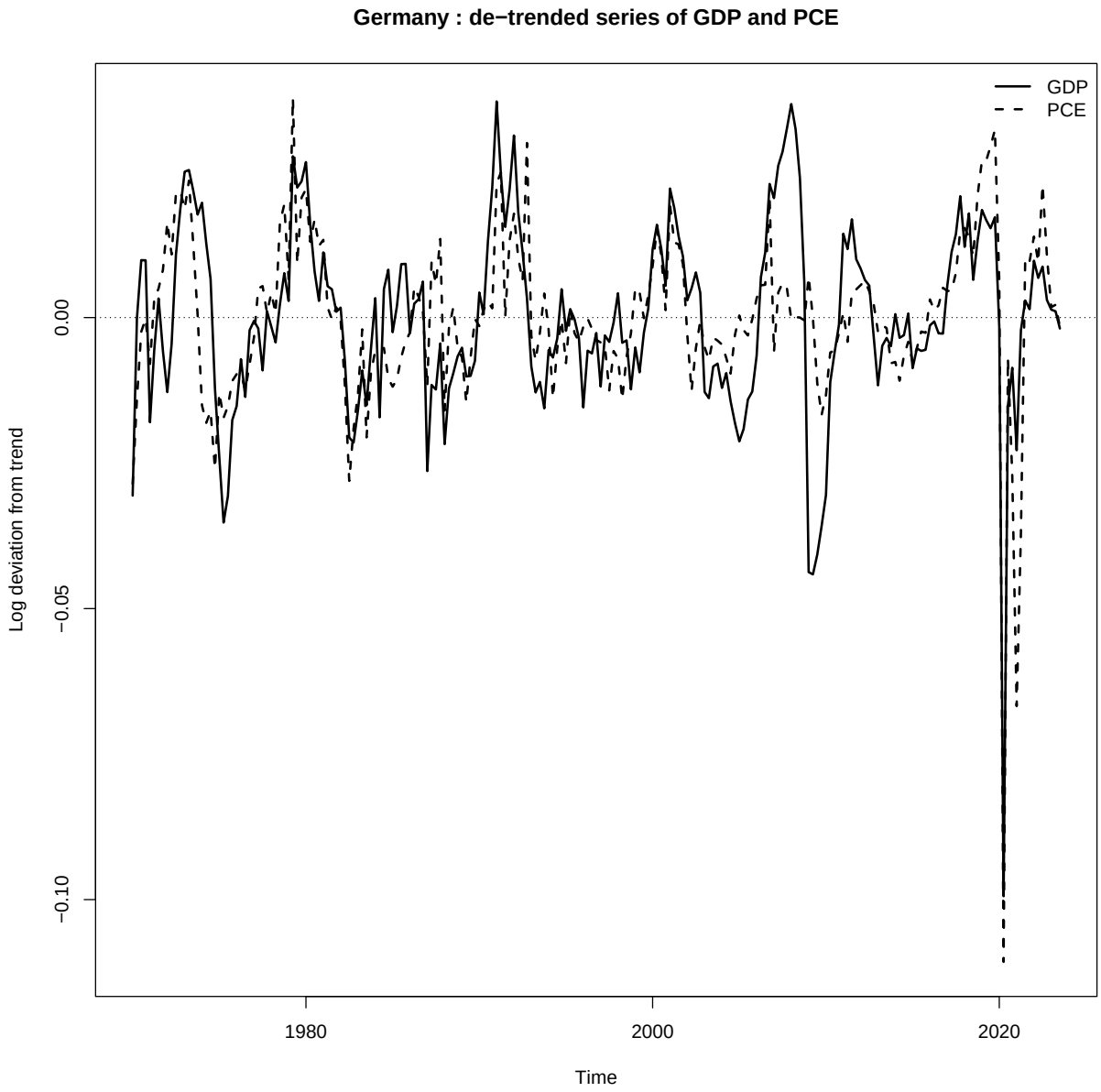

Figure 2: De-trended GDP and PCE in Germany using the HP filter

Task 2

Country	Var(GDP)	SD(GDP)	Var(PCE)	SD(PCE)
United States	0.0002	0.0155	0.0002	0.0143
Germany	0.0003	0.0162	0.0002	0.0146

Table 1: Volatility of detrended GDP and PCE

Country	Corr(GDP,PCE)	Corr(GDP,US GDP)	Corr(PCE,US GDP)
United States	0.8826	–	–
Germany	0.6689	0.5845	0.5275

Table 2: Correlations of detrended GDP and PCE

Task 3

Figure 1 shows the de-trended series for GDP and personal consumption expenditures (PCE) for the U.S., revealing that these move in a fairly similar direction over the observed period. This is also confirmed by the correlation value in Table 2, where the correlation between U.S. detrended GDP and detrended PCE is high and positive (0.8826). Hence, consumption appears to be clearly procyclical in the United States.

Figure 2, which shows the de-trended series for GDP and Private Final Consumption Expenditure for Germany, also shows that both move in a similar direction, although somewhat less strongly than in the USA. This is also shown in Table 2, where the correlation between German detrended GDP and detrended PCE is slightly lower (0.6689), but still positive, suggesting that consumption in Germany also appears to be procyclical. Both figures also show that in times of shocks, such as 2020, both series experience strong negative shocks.

Table 1 shows that the variance and standard deviation of the de-trended German GDP are a little bit higher than those of the U.S. GDP. Furthermore, both German GDP and German consumption correlate positively with the de-trended US GDP (correlations of 0.5845 and 0.5275).

Overall, the results suggest that consumption in both economies is procyclical and that both business cycles exhibit similar patterns and characteristics.

Exercise 2

Task 1

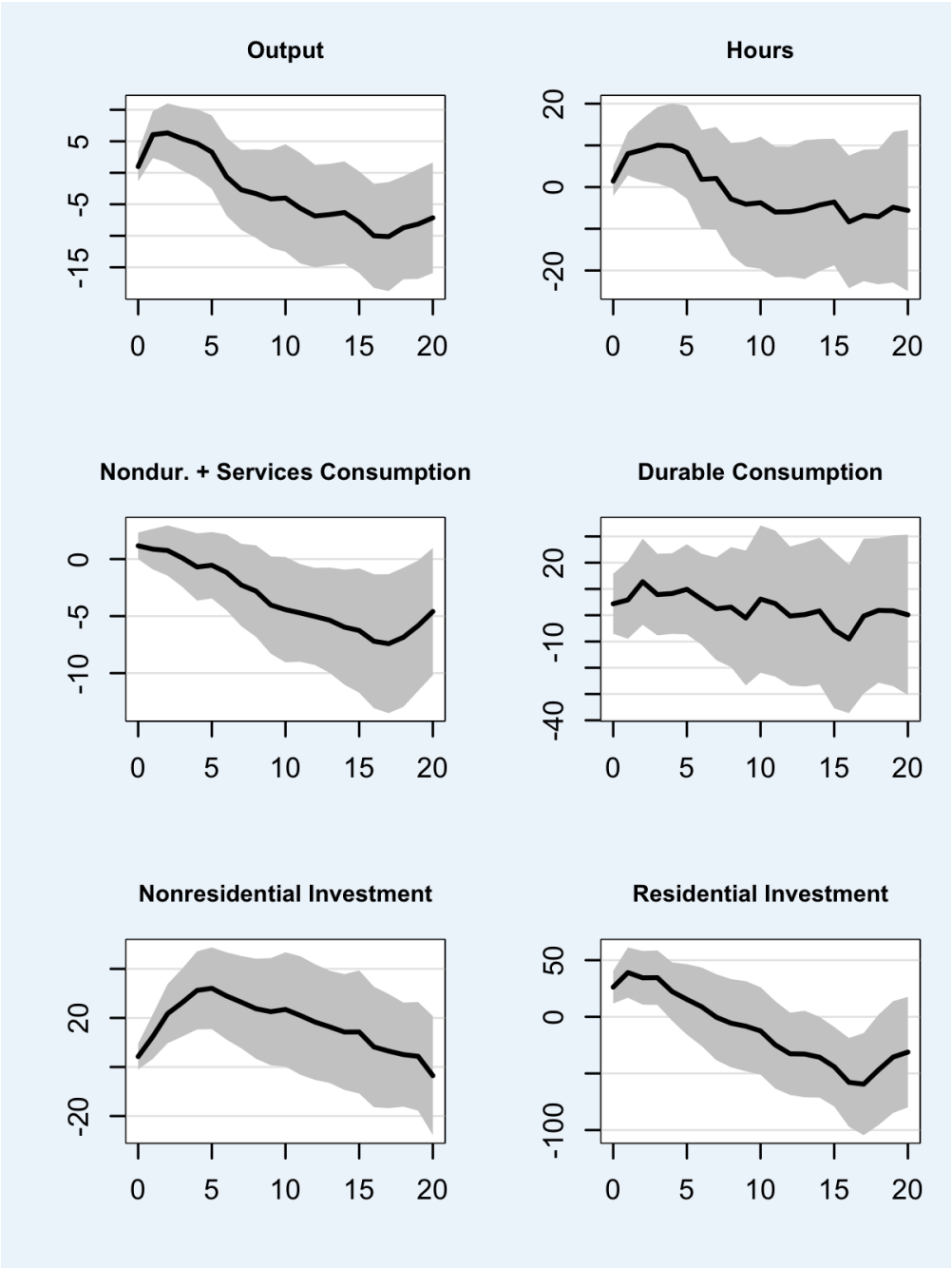

Figure 3: Replication of Figure 8 in Ramey (2016a), based on Data and Code from Ramey (2016b)

Task 2 and Task 3

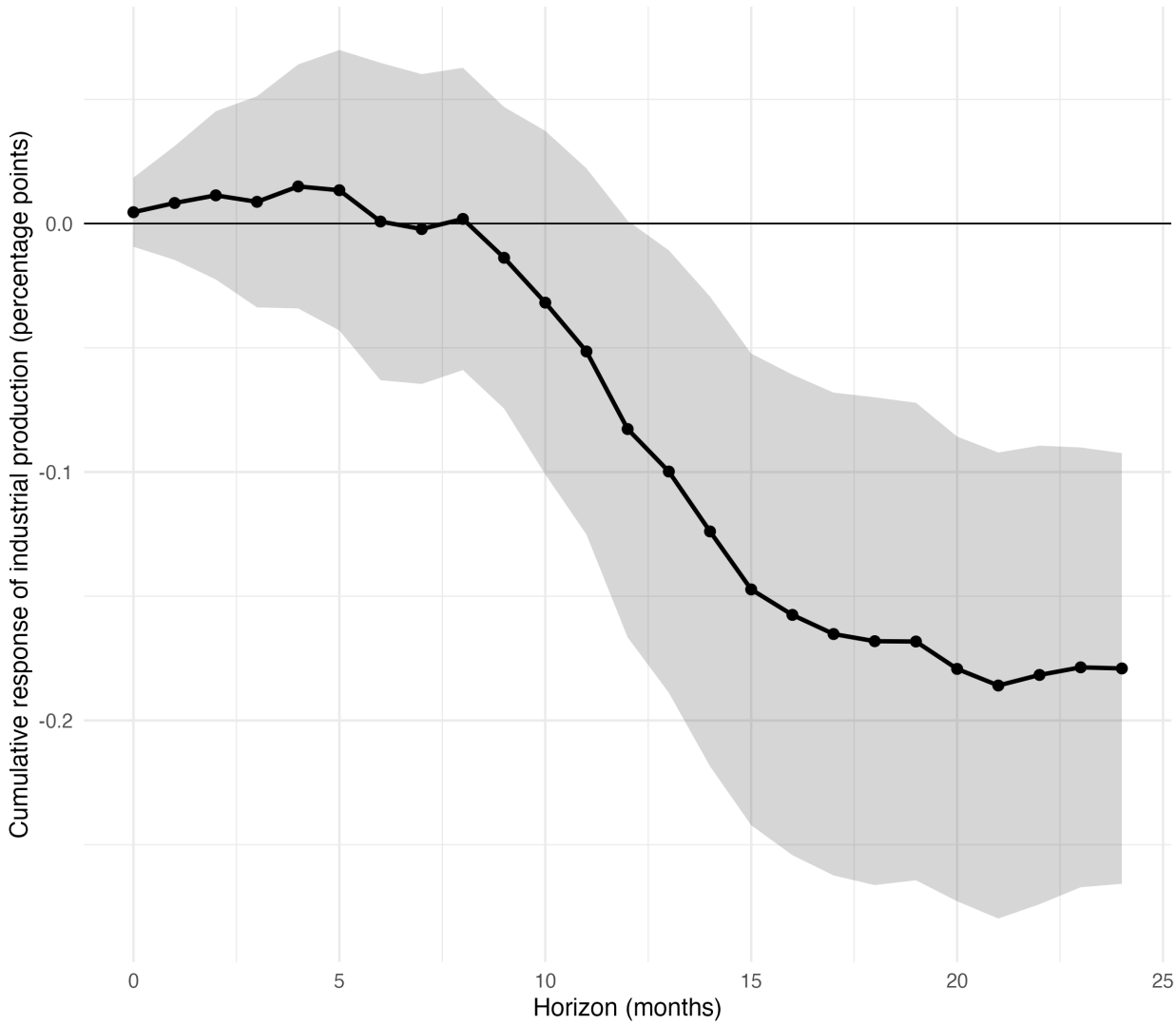

Figure 4: Impulse response of U.S. industrial production (INDPRO) (Board of Governors of the Federal Reserve System (US), 2026) to a shock in the Ramey–Vine real full cost of gasoline (Ramey and Vine, 2012) estimated by local projections. The shock variable is defined as $s_t = 100\,\Delta\log(\text{rv_real}_t)$. The reported response is $100\big(\log(\text{INDPRO}_{t+h}) - \log(\text{INDPRO}_{t-1})\big)$, that is, the cumulative change in log industrial production from month $t-1$ to horizon h. The shaded band represents the 95% confidence interval computed using HAC/Newey–West standard errors. The regression includes 12 lags of industrial production growth and 12 lags of the oil shock as controls.

I estimate the response function of U.S. industrial production to a shock in Ramey–Vine real full cost of gasoline (Ramey and Vine, 2012) using the local projections approach of Jordà (2005). For the oil shock measure I use the Ramey-Vine data available on Valerie Ramey's research page

(Ramey and Vine, 2012). The shock is defined as 100 times the monthly log change in the real full cost of gasoline. I combine this data with the monthly U.S. industrial production index (INDPRO) from FRED (Board of Governors of the Federal Reserve System (US), 2026).

For each horizon $h = 0, \ldots, 24$, I estimate a separate regression, which allows me to analyze the dynamic response of industrial production over time after the shock. The dependent variable, $100(y_{t+h} - y_{t-1})$, where $y_t = \log(\text{INDPRO}_t)$ is the cumulative change in log industrial production from the month $t-1$ to horizon h, expressed in log-point units multiplied by 100. Accordingly, the estimated impulse response function captures the cumulative response.

For each horizon $h = 0, \ldots, 24$, I estimate

$$100(y_{t+h} - y_{t-1}) = \alpha_h + \beta_h s_t + \sum_{i=1}^{12} \gamma_{h,i}\, \Delta y_{t-i} + \sum_{i=1}^{12} \delta_{h,i}\, s_{t-i} + \varepsilon_{t+h},$$

where $y_t = \log(\text{INDPRO}_t)$, $s_t = 100\,\Delta \log(\text{rv_real}_t)$, and $\Delta y_t = 100\,(y_t - y_{t-1})$. Thus, in my regression, β_h is the estimated response of INDPRO to an approximately 1% increase in oil costs in month t. The specification includes 12 lags of monthly industrial production growth and 12 lags of the oil shock as controls. I report 95% confidence intervals based on Newey–West standard errors to account for serial correlation induced by overlapping horizons.

Task 4

The empirical question I examine is the effect of an oil price shock on U.S. industrial production. This question is of interest, since oil price fluctuations are regarded as a significant source of macroeconomic cyclical fluctuations, influencing production costs and household purchasing power. If oil shocks lead to higher energy costs and therefore to higher production cost it can be expected that firms will curtail production.

In Figure 4 one can see that the estimated impulse response function indicates that a positive oil price shock initially has no real effect on production, especially when considering the confidence bands. A decline in production only becomes apparent between the 10th and 15th month after the shock.

This may be because companies react in the short term with other measures, such as initially adjusting working hours or prices before subsequently changing their production level. Alternatively, it's possible that companies already have pre-existing contracts regarding the price of oil and are thus able to maintain their production levels for the first few months after a shock.

In summary, it can be stated that adverse oil price shocks decrease industrial production in the months following the shock, even though the confidence intervals suggest that the strength of the effect should be interpreted with caution and the effect is delayed.

References

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